MAIS PAC

IPhO 02.07: Magnetic Fields and Induction · Worked Solutions

Magnetic Fields and Induction: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Field near a wire and at the centre of a loop. Return →

(a) B = μ_0 I/(2π r) = (4π×10^(-7))(8.0)/(2π×0.050) = 3.2×10^(-5) {T} = 32 μ{T}.

(b) B = μ_0 I/(2R) = (4π×10^(-7))(8.0)/(2×0.040) = 1.26×10^(-4) {T} = 126 μ{T}, about four times larger: the loop concentrates all its current at the same distance, while the wire spreads it along both directions.

(c) n = 400/0.25 = 1600 {m^(-1)}, so B = μ_0 nI = (4π×10^(-7))(1600)(3.0) = 6.0×10^(-3) {T} = 6.0 {mT} — about twice the loop's field, and nearly two hundred times the wire's field at 5 {cm}, because every turn contributes in the same direction. **Check:** all three have the units of μ_0 I/{length}, as they must.

Solution 2 — A proton in a cyclotron. Return →

(a) r = (mv)/(qB) = ((1.67×10^(-27))(5.0×10^6))/((1.60×10^(-19))(0.50)) = 1.04×10^(-1) {m} ≈ 10 {cm}.

(b) T = (2π m)/(qB) = (2π(1.67×10^(-27)))/((1.60×10^(-19))(0.50)) = 1.31×10^(-7) {s}, so f = 1/T = 7.6 {MHz} — the radio-frequency range of a real cyclotron.

(c) K = (1)/(2)mv² = (1)/(2)(1.67×10^(-27))(2.5×10^(13)) = 2.09×10^(-14) {J} = 1.30×10^5 {eV} = 130 {keV}. In T = 2π m/(qB) the speed does not appear at all, so doubling v only doubles r; doubling B halves T and therefore doubles f. **Check:** r/v = 10.4 {cm}/5.0×10^6 {m,s^(-1)} = 21 ns, and 2π times that is 131 ns, consistent with (b).

Solution 3 — A velocity selector and a mass spectrometer. Return →

(a) Undeflected means the electric and magnetic forces cancel: qE = qvB, so

$$v = (E)/(B) = (2.0\times10^{4})/(0.40) = 5.0\times10^{4}\ \{m,s^{-1}\}. \quad (1)$$

This is unusually slow for ions; a practical spectrometer uses a much smaller E.

(b) r = (mv)/(qB) = ((39)(1.66×10^(-27))(5.0×10^4))/((1.60×10^(-19))(0.40)) = (3.24×10^(-24))/(6.4×10^(-20)) = 5.06×10^(-2) {m} ≈ 5.1 {cm}.

(c) With v and B fixed, r ∝ m, so a radius larger by 4.0 {cm} means

$$m = (qBr)/(v) = ((1.60\times10^{-19})(0.40)(9.06\times10^{-2}))/(5.0\times10^{4}) = 1.16\times10^{-25}\ \{kg\} = 69.9\ \{u\} \approx 70\ \{u\}. \quad (2)$$

Isotopes have equal charge but different mass, so equal velocity gives different radii: the beam splits into separate arcs. **Check:** 9.06/5.06 = 1.79 and 70/39 = 1.79 — the ratio of radii equals the ratio of masses.

Solution 4 — Force on a wire and a loop. Return →

(a) F = BI = (0.30)(4.0)(0.25) = 0.30 {N}, perpendicular to both the wire and the field.

(b) A = 5.0×8.0×10^(-4) = 4.0×10^(-3) {m²}, m = NIA = (100)(2.0)(4.0×10^(-3)) = 0.80 {A,m²}, so

$$\tau_{(max)} = mB = (0.80)(0.15) = 0.12\ \{N,m\}. \quad (3)$$

(c) The torque is maximal when the plane of the coil contains the field and falls to zero when the coil's normal is along the field; without a commutator the coil would oscillate about that position, and with one it keeps turning. Real motors are limited by resistive heating in the windings, by the back-emf that grows with speed, and by friction. **Check:** the torque vanishes at the same orientation where the flux through the coil is maximal, since τ = mBsin θ while = BAcos θ.

Solution 5 — Two parallel currents. Return →

(a) Each wire sits in the field of the other, B = μ_0 I/(2π d):

$$(F)/() = ((4\pi\times10\text{^}(-7))(15)(20))/(2\pi(0.040)) = 1.5\times10\text{^}(-3)\ \{N,m\text{^}(-1)\}, \tag{4}$$

and the currents are parallel, so the force is **attractive**.

(b) Between the wires the two fields point in opposite directions. At distance x from the 15 {A} wire,

$$(15)/(x) = (20)/(0.040-x)\ 15(0.040-x) = 20x\ x = (0.60)/(35) = 1.7\times10\text{^}(-2)\ \{m\}, \tag{5}$$

that is 1.7 {cm} from the weaker current.

(c) Antiparallel currents repel with the same magnitude 1.5×10^(-3) {N,m^(-1)}. The net field no longer vanishes anywhere between them (both fields point the same way there), and the wires accelerate apart. **Check:** the neutral point is closer to the smaller current, exactly as for two charges of the same sign in electrostatics.

Solution 6 — A rod on rails (motional emf). Return →

(a) ε = B v = (0.50)(0.40)(3.0) = 0.60 {V}, so I = 0.60/0.60 = 1.0 {A}. The magnetic force on the rod is F = BI = (0.50)(1.0)(0.40) = 0.20 {N}, so an external force of 0.20 {N} must be applied to keep the speed constant.

(b) Mechanical power = Fv = (0.20)(3.0) = 0.60 {W}; electrical power = ε I = I^2R = 0.60 {W}. They balance exactly — the rod is a generator converting mechanical work into heat.

(c) Any current induced by the rod's motion produces a force I× B that opposes the motion: this is Lenz's law in mechanical form, and it guarantees that energy is never created. If the rod is released, that force decelerates it, and all of its kinetic energy ends up as heat in the resistor; it stops once it is at rest, because then ε = 0. **Check:** if the force were assisting the motion, the rod would accelerate and the resistor would heat up with no energy input.

Solution 7 — A rotating coil generator. Return →

(a) ω = 2π(3000/60) = 314 {rad,s^(-1)}, so

$$\varepsilon_0 = NBA\omega = (200)(0.020)(0.12)(314) = 1.5\times10^{2}\ \{V\}. \tag{6}$$

(b) $\varepsilon_{\{rms\}} = \varepsilon_0/\sqrt{2} = 1.5\times10^{2}/1.414 = 1.1\times10^{2}$ {V}.

(c) N = (√(2),ε_({rms}))/(BAω) = (1.414×230)/((0.020)(0.12)(314)) = 431, so about 430 turns.

Check: the emf is sinusoidal because = BAcos ω t and ε = - ∝ sin ω t; the peak occurs when the coil's plane is **parallel** to the field, which is the orientation of maximum flux **change** but zero flux.

Solution 8 — A solenoid as an inductor. Return →

(a) n = 1200/0.20 = 6000 {m^(-1)}, $A = \pi(0.012)^2 = 4.52\times10^{-4}$ {m²}, so

$$L = \mu_0 n^2 A = (4\pi\times10^{-7})(6000)^2(4.52\times10^{-4})(0.20) = 3.6\times10^{-3}\ \{H\} = 3.6\ \{mH\}. \tag{7}$$

(b) $U = \frac{1}{2}LI^2 = \frac{1}{2}(3.6\times10^{-3})(2.5)^2 = 1.1\times10^{-2}$ {J}. The field is $B = \mu_0 nI = (4\pi\times10^{-7})(6000)(2.5) = 1.9\times10^{-2}$ {T}, so

$$u_B = B^2/(2\mu_0) = ((1.88\times10^{-2})^2)/(2(4\pi\times10^{-7}))) = 1.4\times10^{2}\ \{J,m^{-3}\}. \tag{8}$$

Multiplying by the volume $A = 9.05\times10^{-5}$ {m³} returns 1.3×10^{-2} {J} — the same energy, as it must be.

(c) τ = L/R = 3.6×10^(-3)/0.50 = 7.2 {ms}. At the instant of closing, the current is still zero and the whole supply voltage appears across the inductor (L,dI/dt = ε), so the current starts from zero and rises with time constant τ; the resistor's voltage grows only as the current does. **Check:** at t = τ the current has reached 63% of its final value ε/R, the same fraction as a charging capacitor's charge.

Solution 9 — A changing field through a loop. Return →

(a) A = π(0.050)² = 7.85×10^(-3) {m²}, so

$$\varepsilon = A(dB)/(dt) = (7.85\times10^{-3})(0.40) = 3.1\times10^{-3}\ \{V\},\ I = (\varepsilon)/(R) = (3.14\times10^{-3})/(0.020) = 0.16\ \{A\}. \tag{9}$$

(b) If the field points out of the page and grows, the induced current must produce a field **into** the page inside the loop — clockwise, seen from outside. The induced current always opposes the change of flux, not the flux itself: a decreasing field would give the opposite current.

(c) P = I²R = (0.157)²(0.020) = 4.9×10^(-4) {W} = ε²/R. The field energy stored in the loop's volume **increases** as B², so the resistor's heat cannot come from it: the energy comes from whatever external agent is driving the field up, which must do work against the induced current's field. **Check:** ε²/R = (3.14×10^(-3))²/0.02 = 4.9×10^(-4) {W}, matching I²R as energy conservation requires.

Solution 10 — Helical motion and magnetic confinement. Return →

(a) Only the perpendicular component bends the path: v_() = vsin 30° = 2.0×10^6 {m,s^(-1)}, so

$$r = (mv_())/(qB) = ((1.67\times10^{-27})(2.0\times10^{6}))/((1.60\times10^{-19})(0.25)) = 8.4\times10^{-2}\ \{m\},\ T = (2\pi\ m)/(qB) = 2.6\times10^{-7}\ \{s\}. \tag{10}$$

(b) v_() = vcos 30° = 3.46×10^6 {m,s^(-1)}, so the pitch is

$$p = v_()\ T = (3.46\times10^{6})(2.62\times10^{-7}) = 0.91\ \{m\}. \tag{11}$$

(c) The circling charge is itself a tiny current loop, so it carries a magnetic moment μ = (1)/(2)mv_()²/B pointing along - B, and it feels a force -μ, B that pushes it towards **weaker** field. The Lorentz force still does no work on a point charge, so the speed |v| never changes; what the converging field does is turn part of the axial motion into perpendicular motion, so v_() grows and v_() falls while v_()²+v_()² stays fixed. When v_() reaches zero the particle sits at the mirror point, with all its motion perpendicular to the field, and it then unwinds back the way it came. The axial momentum is not destroyed: it is exchanged with the field configuration that produced the converging field, and the particle leaves with the same speed it entered with.

Check: v_()² + v_()² = (2.0² + 3.46²)×10^(12) = 16×10^(12) {m²,s^(-2)} = v², so only the split between the two components changes, never the total speed.